

PROJECT REPORT

Water Quality Test Using Classification Algorithms

1. Introduction

Access to safe drinking water is essential for human health and environmental sustainability. Contaminated water can cause severe health issues, making water quality assessment a critical task. Traditional laboratory-based water testing methods are accurate but time-consuming and expensive.

This project aims to develop a **machine learning-based classification system** to predict whether water is safe for consumption (potable) using physicochemical parameters. By applying classification algorithms, the system provides a fast, automated, and reliable method for water quality assessment.

2. Dataset Description

The dataset used in this project is the **Water Potability Dataset** obtained from Kaggle. It contains **3,276 water samples** with **9 physicochemical attributes** and one target variable.

Target Variable

- **Potability**
 - 0 → Not Safe for Drinking
 - 1 → Safe for Drinking

Input Features

- pH
- Hardness
- Solids
- Chloramines
- Sulfate
- Conductivity
- Organic Carbon
- Trihalomethanes
- Turbidity

The dataset contains missing values in **pH, Sulfate, and Trihalomethanes**, which were handled during preprocessing.

3. Methodology

The project follows a structured machine learning workflow:

3.1 Data Preprocessing

- Missing values were handled using **median imputation**.
- Features were scaled using **StandardScaler**.
- The dataset was split into **training (80%)** and **testing (20%)** sets.
- Class imbalance was analyzed to understand data distribution.

3.2 Exploratory Data Analysis (EDA)

- Distribution plots were used to analyze feature behavior.
- Correlation analysis helped identify relationships between variables.
- Target variable distribution showed imbalance between potable and non-potable water.

3.3 Model Building

The following classification algorithms were applied:

- Logistic Regression
- Decision Tree Classifier
- Random Forest Classifier
- Support Vector Machine (SVM)

Among these, **Random Forest** produced the best overall performance.

4. Results and Evaluation

The Random Forest model achieved the following results:

- **Accuracy:** 67%
- **ROC-AUC Score:** 0.68

Confusion Matrix Analysis

- The model correctly identified most **non-potable water samples**.
- Performance for potable water was comparatively lower due to dataset imbalance.

Classification Metrics

- High recall for non-potable class, which is desirable for public health safety.

- Balanced precision and recall indicate reliable predictions.
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5. Conclusion

This project demonstrates the effective application of machine learning classification algorithms for water quality assessment. The Random Forest classifier showed the best performance with an accuracy of approximately 67% and a ROC-AUC score of 0.68.

Although the dataset is imbalanced, the model performs well in identifying unsafe water, which is crucial for preventing health risks. This system can be used as a **preliminary decision-support tool** for water quality monitoring. Future improvements may include advanced balancing techniques and deep learning models.